

Production-Ready Project Checklist

Purpose:

Building the features is only half the job. A production-ready project should be secure, fast, scalable, user-friendly, accessible, and optimized for deployment. Use this checklist before deploying your project, submitting it for an internship/job, or showcasing it in your portfolio.

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1. Frontend Checklist

Loading States

Users should never wonder whether the application is working.

Include:

- ☐ Skeleton Loaders
- ☐ Spinner (ClipLoader, Circular Progress, etc.)
- ☐ Progress Bar (NProgress or similar)
- ☐ Button Loading State

- ☐ Route/Page Loading
 - ☐ Image Loading Placeholder
-

Error States

Every API request can fail.

Handle:

- ☐ Network Errors
 - ☐ 404 Page
 - ☐ 500 Internal Server Error
 - ☐ Unauthorized (401)
 - ☐ Forbidden (403)
 - ☐ Retry Button
 - ☐ Friendly Error Messages
-

Empty States

Instead of displaying:

No Data Found

Display something meaningful.

Example:

“No notes found yet.
Create your first note.”

Include:

- ☐ Illustration
 - ☐ CTA Button
 - ☐ Helpful Description
-

Form Validation

Every form should validate data.

Frontend

- ☐ Required Fields
- ☐ Email Validation
- ☐ Password Strength
- ☐ Confirm Password
- ☐ Character Limits
- ☐ Live Validation

Backend

- ☐ Validate Again
 - ☐ Never Trust Frontend
-

Debounced Search

Without Debouncing

User types:

H

He

Hel

Hell

Hello

→ 5 API Requests

With Debouncing

User types continuously

→ Only 1 API Request

Checklist

- ☐ Search is Debounced
 - ☐ Delay around 300–500ms
-

Pagination / Infinite Scroll

Never fetch thousands of records at once.

Use:

- ☐ Pagination
 - ☐ Infinite Scroll
 - ☐ Load More Button
-

Lazy Loading

Only load components when required.

Checklist

- ☐ `React.lazy()`
 - ☐ `Suspense`
 - ☐ Lazy Loaded Routes
-

Code Splitting

Large bundles slow websites.

Split:

- ☐ Dashboard
- ☐ Admin Panel
- ☐ Settings
- ☐ Authentication

Image Optimization

Images are usually the largest assets.

Checklist

- ☐ WebP Format
 - ☐ Lazy Loading
 - ☐ Width & Height Defined
 - ☐ Responsive Images
 - ☐ Compressed Images
-

Alt Text

Every meaningful image should include descriptive alt text.

✗ alt="image"

✓ alt="Student learning JavaScript"

Checklist

- ☐ Every Important Image Has Alt Text
-

Responsive Design

Test on

- ☐ Mobile
 - ☐ Tablet
 - ☐ Laptop
 - ☐ Desktop
 - ☐ Large Screens
-

Button States

Buttons should support

- ☐ Hover
 - ☐ Active
 - ☐ Disabled
 - ☐ Loading
-

Notifications

Avoid browser alerts.

Use

- ☐ Sonner
 - ☐ React Hot Toast
 - ☐ Shadcn Toast
-

Theme

- ☐ Light Theme
 - ☐ Dark Theme (Optional)
-

2. Backend Checklist

Authentication

- ☐ JWT
- ☐ Sessions
- ☐ Refresh Tokens

☐ Logout

Authorization

Users should only access what they're allowed to.

- ☐ Admin
 - ☐ User
 - ☐ Moderator
 - ☐ Protected Routes
-

Password Security

- ☐ Password Hashing (bcrypt)
 - ☐ Password Never Stored Plain Text
-

Rate Limiting

Protect APIs from spam.

Checklist

- ☐ Rate Limiter Added
 - ☐ Login Endpoint Protected
 - ☐ Sensitive APIs Protected
-

Input Validation

Never trust user input.

Validate

- ☐ Email

☐ Password

☐ IDs

☐ Numbers

☐ File Uploads

Libraries

- Zod
- Joi
- Express Validator

Error Handling

Return proper HTTP status codes.

☐ 200

☐ 201

☐ 400

☐ 401

☐ 403

☐ 404

☐ 500

Environment Variables

Never upload secrets.

Checklist

☐ .env Used

☐ API Keys Hidden

☐ Database URL Hidden

Logging

Log

- ☐ Errors
 - ☐ API Requests
 - ☐ Important Events
-

CORS

- ☐ Allowed Origins
 - ☐ Allowed Methods
 - ☐ Credentials Configured
-

3. Database Checklist

Indexing

Frequently searched fields should be indexed.

Examples

- ☐ Email
 - ☐ Username
 - ☐ Slug
-

Pagination Queries

Instead of

Find Everything

Use

☐ Skip

☐ Limit

or

☐ Cursor Pagination

Data Validation

☐ Required Fields

☐ Default Values

☐ Enums

☐ Unique Fields

Soft Delete (Optional)

Instead of deleting permanently

Store

deleted = true

4. API Checklist

Every API should have

☐ Validation

☐ Authentication

☐ Authorization

☐ Proper Response Structure

☐ Error Handling

☐ Rate Limiting

☐ Pagination (if required)

Standard Response Format

```
{  
  "success": true,  
  "message": "User fetched successfully",  
  "data": {}  
}
```

5. Performance Checklist

Compression

☐ Gzip

☐ Brotli

Caching

☐ Browser Cache

☐ API Cache

☐ Redis (Optional)

CDN

Store Images on

☐ Cloudinary

☐ ImageKit

☐ Supabase

☐ AWS S3

Lighthouse

Aim for

- ☐ Performance
 - ☐ Accessibility
 - ☐ SEO
 - ☐ Best Practices
-

Bundle Optimization

- ☐ Tree Shaking
 - ☐ Code Splitting
 - ☐ Lazy Loading
 - ☐ Remove Unused Packages
-

6. SEO Checklist

Metadata

- ☐ Page Title
 - ☐ Meta Description
 - ☐ Keywords (Optional)
-

Open Graph

- ☐ og:title
- ☐ og:description

☐ og:image

☐ og:url

Twitter Cards

☐ Twitter Title

☐ Twitter Image

☐ Twitter Description

Technical SEO

☐ robots.txt

☐ sitemap.xml

☐ Canonical URLs

☐ favicon

☐ Structured Data (Schema.org)

HTML Structure

☐ One H1

☐ Proper H2

☐ Proper H3

☐ Semantic HTML

<header>

<main>

<section>

<footer>

7. Security Checklist

General

- ☐ HTTPS
 - ☐ Helmet
 - ☐ Secure Headers
-

Protection

- ☐ XSS Protection
 - ☐ SQL Injection Prevention
 - ☐ NoSQL Injection Prevention
 - ☐ CSRF Protection (if using cookies)
-

Cookies

- ☐ HttpOnly
 - ☐ Secure
 - ☐ SameSite
-

Upload Security

- ☐ File Size Limit
 - ☐ MIME Type Validation
 - ☐ Allowed Extensions
-

8. User Experience Checklist

- ☐ Copy to Clipboard
 - ☐ Confirmation Dialog Before Delete
 - ☐ Undo Button
 - ☐ Search
 - ☐ Filters
 - ☐ Sorting
 - ☐ Persistent Login
 - ☐ Remember User Preferences
 - ☐ Smooth Navigation
 - ☐ Helpful Empty States
 - ☐ Friendly Error Messages
-

9. Accessibility Checklist

- ☐ Semantic HTML
 - ☐ Keyboard Navigation
 - ☐ Focus Indicators
 - ☐ Labels for Inputs
 - ☐ ARIA Attributes
 - ☐ Proper Color Contrast
 - ☐ Screen Reader Friendly
 - ☐ Descriptive Alt Text
-

10. Analytics & Monitoring

- ☐ Google Analytics
 - ☐ Google Search Console
 - ☐ Error Tracking (Sentry)
 - ☐ API Logs
 - ☐ Performance Monitoring
-

11. Deployment Checklist

Before Deploying

- ☐ Production Build Works
 - ☐ Environment Variables Added
 - ☐ Database Connected
 - ☐ SSL Certificate Active
 - ☐ Custom Domain Connected
 - ☐ robots.txt Accessible
 - ☐ sitemap.xml Generated
 - ☐ Favicon Working
 - ☐ Manifest File Added (PWA if applicable)
 - ☐ No Console Errors
 - ☐ No Network Errors
 - ☐ Images Loading Properly
 - ☐ All Routes Working
 - ☐ 404 Page Exists
 - ☐ Redirects Working
-

12. Testing Checklist

Manual Testing

- ☐ Login
 - ☐ Signup
 - ☐ Logout
 - ☐ CRUD Operations
 - ☐ File Upload
 - ☐ Search
 - ☐ Pagination
 - ☐ Responsive Layout
-

Browser Testing

- ☐ Chrome
 - ☐ Firefox
 - ☐ Edge
 - ☐ Safari
-

Device Testing

- ☐ Android
 - ☐ iPhone
 - ☐ Tablet
 - ☐ Desktop
-

13. Final Resume Checklist

Before adding your project to your resume, ask yourself:

- ☐ Is the project deployed?
 - ☐ Does every feature actually work?
 - ☐ Is it fully responsive?
 - ☐ Does it have loading, error, and empty states?
 - ☐ Is the backend secure?
 - ☐ Is rate limiting implemented?
 - ☐ Are API responses consistent?
 - ☐ Are forms validated?
 - ☐ Is SEO configured?
 - ☐ Is performance optimized?
 - ☐ Is accessibility considered?
 - ☐ Is the project scalable?
 - ☐ Is the code clean and modular?
 - ☐ Is the README complete?
 - ☐ Have you tested it on multiple browsers?
 - ☐ Would you confidently show this project in a technical interview?
-

Production Ready Scorecard

Give yourself **1 point** for every completed item.

- **90%+** →  Production Ready
- **75–89%** →  Good, but needs improvement
- **50–74%** →  Decent college project, not production-ready
- **Below 50%** →  Focus on improving quality before showcasing it